Megan Tjandrasuwita

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Education

Massachusetts Institute of Technology 2022 - Present
Ph.D. in Computer Science, GPA 5.0/5.0

California Institute of Technology 2018 - 2022
B.S. in Computer Science, GPA: 4.0/4.0

Research Interests

Enhancing the emergence of human-like perception and reasoning in multimodal foundation models. Aligning vision, text, and code modalities to enable visual reasoning, such as sketches and image generation. Code generation to improve reasoning.

Skills

Multimodal learning • Large vision language models • Representation similarity analysis • Deep learning architectures

Selected Publications

1. **Megan Tjandrasuwita**, Paul Pu Liang, Armando Solar-Lezama. **ChartRef: Benchmarking Fine-Grained Visual Element Localization in Charts.** *In submission to ICLR 2026.*
2. Hengzhi Li, (equal contribution first authors), **Megan Tjandrasuwita** (2nd author), ... **PuzzleWorld: A Benchmark for Multimodal, Open-Ended Reasoning in Puzzle Hunts.** *In submission to ICLR 2026.* [paper]
3. **Megan Tjandrasuwita**, Chanakya Ekbote, Liu Ziyin, Paul Pu Liang. **Understanding the Emergence of Multimodal Representation Alignment.** *ICML 2025, ICLR 2025 Re-Align Workshop (oral).* [paper]
4. Hengzhi Li, **Megan Tjandrasuwita**, Yi R. Fung, Armando Solar-Lezama, Paul Pu Liang. **Towards Socially-Intelligent Nonverbal Foundation Models.** *NeurIPS 2025.* [paper]
5. **Megan Tjandrasuwita**, Jie Xu, Armando Solar-Lezama, Wojciech Matusik. **MeMo: Meaningful, Modular Controllers via Noise Injection.** *NeurIPS 2024.* [paper]
6. Liane Makatura, Michael Foshey, ..., **Megan Tjandrasuwita** (7th author), ... **How Can Large Language Models Help Humans in Design and Manufacturing?** *Harvard Data Science Review 2024.* [*part1*, *part2*]
7. Tailin Wu, **Megan Tjandrasuwita**, Zhengxuan Wu, Xuelin Yang, Kevin Liu, Rok Sosic, Jure Leskovec. **ZeroC: A Neuro-Symbolic Model for Zero-shot Concept Recognition and Acquisition at Inference Time**. *NeurIPS 2022, ICML 2022 Beyond Bayes Workshop.* [*paper*]
8. **Megan Tjandrasuwita**, Jennifer J. Sun, Ann Kennedy, Swarat Chaudhuri, Yisong Yue. **Interpreting Expert Annotation Differences in Animal Behavior**. *CVPR 2021 CV4Animals Workshop.* [*paper*]

Research Experience

Multisensory Intelligence Lab Oct 2024 - Present
Deep Learning Researcher *Advisor: Professor Paul Liang*

- Enhance spatial reasoning and chart understanding capabilities of multimodal foundation models.

- Experiment with vision foundation models and LLM post-training (SFT and RL) to enhance fine-grained visual grounding in charts. Conference paper in submission to *ICLR 2026*.
- Extensively evaluate the emergence of alignment between unimodal encoders across varying levels of redundant information and heterogeneity between modalities. Conference paper published at *ICML 2025*.
- Curate benchmarks on reasoning about open-ended puzzles (In submission to *ICLR 2026)* and understanding videos of mime performances that lack dialogue (*NeurIPS 2025*).

Computer-Aided Programming Group — Sep 2022 - Present
Deep Learning Researcher — *Advisor: Professor Armando Solar-Lezama*

- Developed ML pipeline involving imitation learning for pretraining neural network modules and reinforcement learning for transferring modules to different structures and tasks. Conference paper published at *NeurIPS 2024*.

Stanford Network Analysis Project, Stanford — Jun 2021 - Jun 2022
Deep Learning Researcher — *Advisor: Professor Jure Leskovec*

- Researched neurosymbolic methods for performing human-like concept recognition and reasoning in visual domains. Conference paper published at *NeurIPS 2022, ICML 2022 Beyond Bayes Workshop*.

Machine Learning Group, Caltech — Sep 2020 - Jun 2022
Machine Learning Researcher — *Advisor: Professor Yisong Yue*

- Employed program synthesis to generate human-interpretable programs that classify animal behavior based on laboratory datasets. First-authored paper published at the *CVPR 2021 CV4Animals Workshop*.

Industry Experience

Microsoft — Oct 2025 - Present
Research Intern — *Mentor: Wenshan Wu*

- Research multimodal in-context learning.

Oracle - Corporate Architecture — Jun 2020 - Sep 2020
Software Engineer Intern

- Applied machine learning algorithms to create a recommendation system that processes purchase requests from Oracle's employees. Achieved highly interpretable outputs through its ranking system and had a precision of over 80% on validation examples.

Teaching Experience

Caltech Teaching Assistant — Jan 2022 - Jun 2022

- Held office hours, graded problem sets, and prepared lecture notes for graduate-level computer science courses: "Machine Learning and Data Mining" (CS/EE 155) and "Advanced Machine Learning Methods" (CS/EE 159).

Academic Service

Reviewer for *NeurIPS 2025, ICML 2025*

Honors & Awards

NSF Graduate Research Fellowship (GRFP) — 2022
MIT Stata Family Presidential Fellowship — 2022